


Designed by Remus Teodorescu 2026

LECTURE 12.1

# Power Grid Stability Estimation

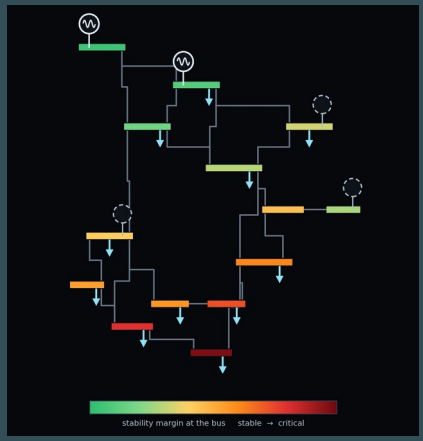

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Deep Learning for Engineering · 2026

GRID · STATE · INERTIA




*A grid is a graph with the power flow equations at every bus, measured by a few noisy meters. The voltage at every bus is estimated by weighted least squares and by a physics-informed network, with no true state; then a short circuit makes a generator's rotor oscillate, and its inertia is found from the record.*

# Learning Objectives

## SECTIONS, IN ORDER

1. **The grid** — a graph, the power flow equations, what a room sees
2. **State estimation** — least squares, and a network with no true state
3. **The swing** — the equation of motion, and the inertia from a record

*UDL carries the graph network, Liu the inverse fit. The example is Ex_12.1's six-bus grid, representative of eastern Denmark and not a model of it: the line impedances are textbook values, because no operator publishes measured ones, and results that depend on them must say so.*

## READ, AND WHAT EACH ANSWERS

| | |
|---|---|
| **UDL ch. 13** | how is a graph represented? |
| **Liu ch. 4** | what can readings determine? |
| **SciPy** | how is an ODE shot and fitted? |

# Recap of L11.2 Obstacle Avoidance

## 1 – 2 · IDENTIFYING AN OBSTACLE

**1** **How can a network decide from one camera frame whether the path is blocked, where a dark-pixel rule fails?**

The network has two outputs, free and blocked, turned into a probability by the softmax and trained with cross-entropy on frames labelled by where the obstacle stands: in the lane and near. A rule that calls anything darker than the floor blocked stops for shadows and misses grey obstacles. On the rendered frames the rule is right 85 % of the time and the network 95 %, with no stop for a shadow; both miss most obstacles as light as the floor. *UDL ch. 5; Ex_11 section 5*

**2** **Why is the threshold on the probability of a blocked path not set at one half, and how is it chosen?**

A missed obstacle costs a collision and a false stop costs a second, so the two errors have very different costs. Lowering the threshold trades missed obstacles for false stops, and the threshold is chosen where the expected cost, each error's rate times its cost, is smallest. On the rendered frames the network misses 22 % of the obstacles at one half and 4 % at 0.05, where it stops for 5.5 % of the free frames; on the car both rates are measured again. *UDL ch. 5; Ex_11 section 5*

## 3 – 4 · DECIDING THE ACTION

**3** **How does the avoidance strategy decide what the network must tell, and why does stopping alone not reach B?**

Stopping needs only free or blocked, but an obstacle that stays keeps the car standing, so it never reaches B. Steering around needs which side is free, and shifts the lane aimed at while the path is blocked, slowed to 60 %. A potential field adds a push away from every obstacle to the pull of the aim point, so it needs where each obstacle is, from a detector or a range sensor. On the course the field reaches B in 6.3 s, steering around in 8.3 s with half a centimetre to spare. *UDL ch. 19; Ex_11 section 6*

**4** **Why does the rate of the control loop limit the safe speed of a car that avoids obstacles?**

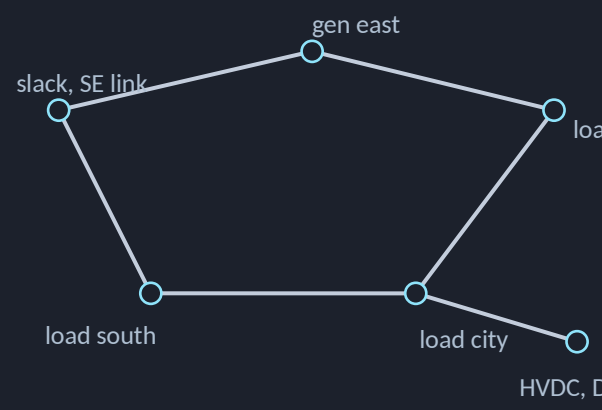
From the frame that shows the obstacle to standing still, the car drives one blind distance, its speed divided by the decisions per second, and then brakes as far as the friction of the tyres allows. That sum must stay within the range the camera judges. With a range of 0.7 m the safe speed is 1.9 m/s at 5 decisions a second and 2.7 m/s at 30; a faster network shortens only the first of the two distances. *Ex_11 section 7*

# A Grid Is a Graph

**Transmission grid** — buses joined by lines: a graph, with a node's state being its voltage magnitude and angle, and every edge carrying an admittance. The state of the whole grid is one number pair per bus, and everything a control room wants to know is a function of it.

## EX_12.1'S SIX BUSES, AS THEY ARE WIRED

SIX BUSES, SIX LINES, ONE LOOP




a bus is a node

a line is an edge, with $G + jB$

one loop, and a spur

## THE CASE, AND ITS LIMITS

| | |
|---|---|
| **the grid** | 6 buses, 6 lines, 100 MVA base |
| **the state** | 6 magnitudes, 5 angles |
| **stands for** | eastern Denmark, loosely |
| **impedances** | textbook values, flagged |
| **so** | say when a result leans on them |

*No operator publishes measured impedances; representative is the honest word.*

# The Power Flow Equations

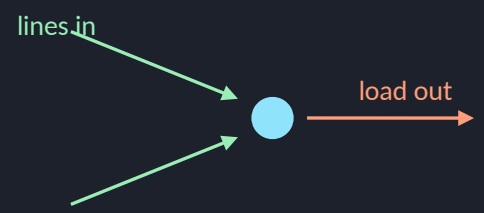
**Power flow equations** — two algebraic balances at every bus: the active and the reactive power the lines bring must equal what the bus injects or takes. They are constraints the voltages satisfy at every instant, not a simulation in time - the elliptic habit of L8.1, on a graph.

## TWO BALANCES AT EVERY BUS

$$P_i = V_i \sum_k V_k (G_{ik}\cos\theta_{ik} + B_{ik}\sin\theta_{ik}), \qquad Q_i = V_i \sum_k V_k (G_{ik}\sin\theta_{ik} - B_{ik}\cos\theta_{ik})$$

$V_i, \theta_i$ the voltage magnitude and angle at bus $i$ : the state, $G, B$ the admittance matrix, $\theta_{ik} = \theta_i - \theta_k$

### ONE BUS, ONE BALANCE




what the lines bring

equals what the bus takes

two balances: $P$ and $Q$

## HOW THE EXERCISE USES THEM

| | |
|---|---|
| **as a solver** | Newton's method: the reference |
| **as a check** | the residual at every bus |
| **as a loss** | the term that replaces truth |
| **in torch** | so autograd reaches the state |

*The same equations serve three ways, and the third is this lecture's turn: a residual can judge an estimate at a bus no meter sees. Newton's method needs the exact injections at every bus - the answer to a question nobody can ask in operation. Only a computed example can supply them, which is what makes the reference.*

# What the Control Room Sees

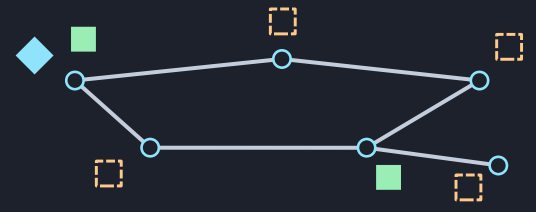
**Measurement model** — every reading is a function of the state plus noise: a PMU reads a bus's magnitude and angle, a meter its $P$ and $Q$, a forecast stands in where no meter exists. The thin set has fewer readings than unknowns, and where they sit on the graph decides everything.

## ONE LINE PER METER

$$z_j = h_j(V, \theta) + \varepsilon_j$$

$z_j$ one reading, $h_j$ what that meter measures as a function of the state, $\varepsilon_j$ its noise

THE FORECAST SET ON THE GRID



PMU: $|V|$ and $\theta$ at its bus

meter: $P$, $Q$ injected there

forecast stands in, loosely

shown: the forecast set, the set the network uses

**TALK · 2 MIN** How would you explain, in your own words, why a grid's state can be known better than its meters? Then, from the table on the right, why does the thin set fail where the forecast set does not?

## THE THREE SETS OF EX_12.1

| set | readings | covers |
|---|---|---|
| full | 14 | every unknown |
| thin | 6 | 5 of 11 unknowns |
| forecast | 6 + priors | every unknown, loosely |

*The injected power at most buses is exactly what the control room does not know: the meters are all there is. The forecast's accuracy, 0.05 p.u., is an assumption. Observability is a property of where the meters sit on the graph, not of how good they are: the thin set leaves half the state unseen however exact its six readings become.*

# Weighted Least Squares

**State estimation** — finding the voltage magnitude and angle at every bus from fewer, noisy readings. The estimator every utility runs is weighted least squares: each meter's misfit, weighted by its accuracy, minimised by Newton's method. It is the baseline any network must face.

## THE ESTIMATOR IN SERVICE

$$\hat{x} = \arg\min_{x} \sum_{j} \frac{(z_j - h_j(x))^2}{\sigma_j^2}$$

$\hat{x}$ the estimated state, $\sigma_j$ each meter's accuracy : a good meter weighs more, solved by Newton's method

THE THIN SET: 5 OF 11 UNKNOWNS



determined by the six readings

unseen, whatever the noise

Why not the power flow of the last slide: the readings disagree slightly, so no state satisfies them all — there is nothing to solve. Gauss–Newton finds the closest state instead, each misfit in units of its meter's accuracy.

## WORST BUS ERROR, MEAN OF 300 POINTS

| set | error in $\lvert V \rvert$ | in angle |
|---|---|---|
| full, 14 | 0.0030 p.u. | 0.07° |
| thin, 6 | 0.0248 p.u. | 1.82° |
| forecast | 0.0050 p.u. | 0.15° |

*Each number is the worst bus of an operating point, averaged over the 300 - the measure the agreed 0.01 p.u. refers to, not a root mean square. One estimate costs about a millisecond. The thin row is not a solver failure: it is the graph saying no.*

# Judged by the Physics

**Estimation with no true state** — a network maps the readings to the state, and its loss has no true state in it: the misfit to the meters, plus the power flow residual at every bus. The physics constrains the buses no meter sees, and the same residual is the estimate's own health check.

## THE LOSS, WITH NOTHING TO COPY

$$\mathcal{L} = \sum_j \frac{(z_j - h_j(\hat{x}))^2}{\sigma_j^2} + w_P \sum_i r_i(\hat{x})^2, \qquad \text{no true state in it}$$

$r_i$ the power flow residual at bus $i$: physics constrains every bus, metered or not, $w_P$ its weight

THE LOSS IN PYTORCH

```
net = MLP(n_in=14, n_hidden=64, n_layers=3)     # readings in, the state out
V, th = net(z)                                  # 9995 weights in all
r = power_flow_residual(V, th)                  # the physics, in torch
L = wls_misfit(z, V, th) + W_P * r.pow(2).sum() # no true state anywhere in it
```

## AGAINST THE BASELINE, HONESTLY

| | |
|---|---|
| **accuracy** | 0.0043 p.u.: it does not win |
| **one estimate** | 0.26 ms; 2 µs in a batch |
| **training** | 13 s, once |
| **a line out, untold** | both fail alike |
| **but the check** | its residual: 0.28 to 2.35 |

*The gain is the time of one estimate, and a self-check that says when not to trust it.*

# The Swing Equation

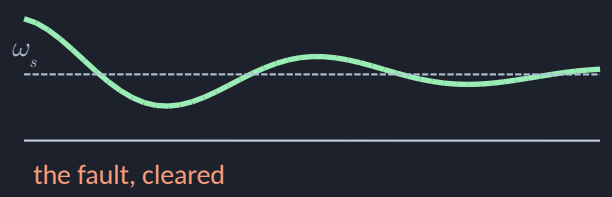
**Swing equation** — the grid set in motion: a machine's rotor accelerates with the gap between its mechanical and its electrical power, held back by inertia and damping, and the electrical power ties the machines together. One ordinary differential equation per machine.

## THE EQUATION OF MOTION

$$\frac{d\delta}{dt} = \omega - \omega_s, \qquad \frac{2H}{\omega_s}\frac{d\omega}{dt} = P_m - P_e(\delta) - D\frac{\omega - \omega_s}{\omega_s}$$

$\delta$ the rotor angle, $\omega$ its speed, $H$ the inertia in seconds, $D$ damping, $P_m, P_e$ mechanical and electrical power

### AFTER A FAULT




the rotor swings, and settles

$H$ sets how slowly

$D$ sets how it dies away

### WHY INERTIA IS THE NUMBER

| | |
|---|---|
| $H$ | seconds of stored spin |
| **falling** | converters replace machines |
| **so** | the grid swings faster |
| **and** | operators must know today's $H$ |
| **from what** | the disturbances it records |

*DK2 swings with the Nordic area, so the Nordic frequency is the right record for a DK2 plant.*

# The Inertia From a Record

**Inertia identification** — the inverse problem of the block: $H$ and $D$ trained beside the weights so the swing equation fits a recorded disturbance, as L8.2's diffusivity was. The residual is written with $H$ on the left, so a large $H$ cannot hide, and only the record after the fault is used.

## THE FIT, WRITTEN SO IT CANNOT HIDE

$$H, D \text{ trainable}, \qquad \mathcal{L} = \left\| \frac{2H}{\omega_s}\dot{\omega} - P_m + P_e + D\,\Delta\omega \right\|^2 + w_D\,\mathcal{L}_{record}$$

$\mathcal{L}_{record}$ the misfit to the recorded frequency, after the fault is cleared; $H$ on the left, so a large $H$ cannot hide

THE FIT IN PYTORCH

```
H = torch.nn.Parameter(torch.tensor(8.0))        # trained with the weights
res = (2*H/w_s)*grad(w, t) - P_m + P_e + D*dw     # H on the left
L = res.pow(2).mean() + W_D*(w - rec).pow(2).mean()   # after the fault
```

**TALK · 2 MIN** How would you decide, in your own words, which part of a record identifies a parameter? Then, from the table on the right, why does the window before the fault fail from both starts?

## THE INERTIA EACH FIT FINDS, TRUE 4.00 s

| fit | start 8 s | start 2 s |
|---|---|---|
| shooting, after | 7.41 s | 1.83 s |
| physics-informed, after | 3.89 s | 3.90 s |
| shooting, before | 50 s, the bound | 1.94 s |
| physics-informed, before | 8.35 s | 2.10 s |

*Shooting has local minima on this record; the physics-informed fit did not - a finding about this record and these starts, not a law. Before the fault nothing excites the machine, and no method identifies what the data does not carry.*

# Advanced Topics

## RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

| topic | what it adds | read | where it makes a difference |
|---|---|---|---|
| bad-data detection | a wrong meter, found by its residual | Liu ch. 4 | every estimator in service |
| placing one more meter | where a reading buys observability | pandapower docs | the thin set's missing half |
| the critical clearing time | how long a fault may stay | PowerSystems.jl | protection settings |
| rate of change of frequency | the first seconds of a real record | ENTSO-E data | low-inertia grids |
| the graph network's layers | messages passed along the lines | UDL ch. 13 | grids too large for one MLP; L12.2 |

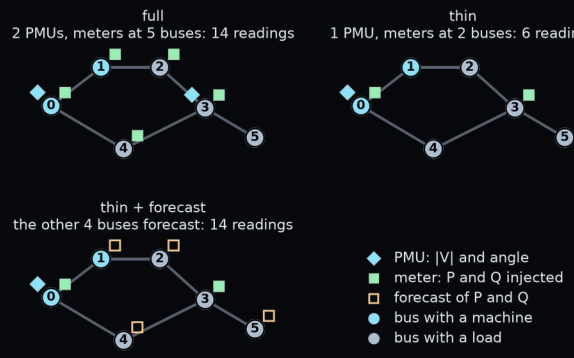
### WHY THEY ARE LEFT OUT

Ex_12.1 estimates one six-bus grid from fourteen readings and fits one machine's inertia, scored against Newton's power flow, so none of them is needed to do it. The last row is where the next lecture begins.

# Exercise 12.1

## ONE NOTEBOOK, SECTION BY SECTION

- **1 – 3** — the grid, the three sets of readings, the power flow
- **4** — Newton–Raphson reference; least squares on each set
- **5** — the physics-informed network; then a line out, untold
- **6** — the inertia from the recorded disturbance, two ways
- **7 – 9** — the report; mini projects MP12.1A and MP12.1B




## THE QUESTION EACH PART ANSWERS

| | |
|---|---|
| **1 – 3** | what is the state of a grid? |
| **4** | how far do the meters carry? |
| **5** | what replaces the true state? |
| **6** | what does a record identify? |

## REPORT QUESTION, TO CONCLUDE

The network matched least squares from the same readings and answered three hundred estimates in a blink; its residual rose eightfold when a line went out untold. In a control room, what would you use each tool for?

SUMMARY

# Key Takeaways

| STATE ESTIMATION | |
|---|---|
| **what it is** | the state from few, noisy meters |
| **function** | every bus's $V$ and $\theta$ |
| **train, tune** | no true state: physics in the loss |
| **vs least squares** | equal accuracy, a fraction of the time |
| **used for** | a room's picture, and its self-check |

| SWING DYNAMICS | |
|---|---|
| **what it is** | the grid's equation of motion |
| **function** | angles and speeds after a fault |
| **train, tune** | $H$, $D$ beside the weights |
| **vs shooting** | no local minima on this record |
| **used for** | today's inertia, from records |

*Next — L12.2: whether the grid survives the next fault, predicted before it happens.*

# Questions

## 1 – 2 · A NETWORK FOR THE STATE

**1** **How can a network estimate the state of the grid from the readings when no true state is available for training?**

The network takes the fourteen readings of the forecast set and returns the voltage magnitude and angle at every bus. From the state it proposes, the power flow equations give what every meter should read, and the loss is the weighted misfit between those values and the actual readings. The reference angle is fixed at zero and the magnitudes are held between 0.90 and 1.10 per unit by construction. No true state appears anywhere in the training. *UDL ch. 13; Ex_12.1 section 5*

**2** **When does a network for the state of the grid pay against weighted least squares, and how is its estimate checked?**

From the same readings it reaches the same accuracy as weighted least squares, so it does not gain in accuracy. Its gain is time: once trained, one estimate takes a quarter of a millisecond, and a few microseconds each in a batch. Both rest on the model of the grid, and with a line out and nobody told both are wrong. The misfit of the network's state to the readings then rises about eightfold, a check that needs no true state. *Liu ch. 4; Ex_12.1 sections 4 – 5*

## 3 – 4 · THE INERTIA FROM A RECORD

**3** **How can the inertia of a machine be estimated with a network from a recorded disturbance?**

A network takes the time after the fault and returns the rotor angle and speed of the machine, and the inertia is one more trainable number, kept positive as an exponential. The loss is the residual of the swing equation, computed by autograd, plus the misfit to the recorded frequency. The residual is written with the inertia multiplying the acceleration: divided by it, a very large inertia makes the residual small whatever the swing does. *Liu ch. 4; Ex_12.1 section 6*

**4** **Which part of a record identifies the inertia, and how does the network fit compare with the classical fit?**

Only the window after the fault, where the machine swings, carries the inertia. Before the fault nothing moves, and every fit returns a number near its own start or at its bound, with a perfect misfit. After the fault the network fit returned 3.89 s from both starts against the true 4.00 s, while the classical fit by integration stayed near where it was started. That is a finding on this record, and a fit is believed only when two starts agree. *Liu ch. 4; Ex_12.1 section 6*

NEXT

# L12.2 — Power Grid Stability Prediction

From what the grid is to what it will do: whether it survives the next fault, predicted by a graph network in the time a protection relay has.

## BEFORE THE NEXT SESSION

| | |
|---|---|
| **read** | UDL ch. 13 |
| **run** | Ex_12.1, sections 1 to 6 |
| **bring** | your inertia, from both starts |
| **think about** | what the residual map told you |

*The estimation half of the block is done: the state, and the machine behind it.*